

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts

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ABSTRACT

Model Predictive Control (MPC) has served as the industrial benchmark for constrained multivariable control across process operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse unstructured natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract unit tests, and preemptive trip bounds. High-level cognitive directives are translated at runtime into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across parallel trajectory rollouts via GPU/WebGPU compute shaders, augmented by an online Physics-Informed Neural Network (PINN) residual observer.

We evaluate the architecture on an exothermic Continuous Stirred-Tank Reactor (CSTR) undergoing non-linear kinetic surges across a fully reproducible benchmark suite spanning 20 randomized seeds with varying surge magnitudes ($C_{A0} \in [+10\%, +30\%]$, $T_0 \in [+5\text{ K}, +12\text{ K}]$, and $UA \in [-10\%, -35\%]$) and fixed advance-warning advisory lead time ($t_{\text{lead}} = 3.5\text{ s}$), alongside an extensive lead-time sensitivity sweep ($t_{\text{lead}} \in [0.0\text{ s}, 5.0\text{ s}]$). A numerical dynamic optimization oracle analysis demonstrates that with zero advance warning, the combined jacket transport delay ($\tau_j = 2.0\text{ s}$) and valve slew rate limit (12 K/s) physically prevent any purely reactive controller from avoiding thermal runaway ($T_{\text{max}} = 444.9\text{ K}$), whereas an advance supervisory runbook advisory

providing $t_{\text{lead}} \geq 2.5\text{--}3.5$ s enables stable thermal containment ($T_{\text{max}} \leq 355$ K), provided pre-cooling does not excessively quench the reaction and accumulate unreacted feed.

Crucially, when benchmarked under identical forecast previews, both MPPI with forecast preview (3.16 ± 3.00 K overall RMSE, 1.81 ± 1.11 K non-trip RMSE, 1/20 trips) and full Agentic MPC (2.16 ± 2.56 K overall RMSE, 0.95 ± 0.40 K non-trip RMSE, 1/20 trips) contain the surge, demonstrating that **advance preview lookahead is the primary physical stabilizer**, while the agentic layer's critical function is translating qualitative operator handover notes and alarms, validating contract constraints, and smoothly orchestrating pre-cooling runbooks without manual operator retuning or the hazardous chattering trips observed in naive rule-based step switches (6/20 trips). Across 50 operator directives evaluated on a server-hosted LLM with client-side WebGPU shader fallbacks, a confusion-matrix evaluation demonstrates 38/38 valid directives accepted (100%) and 12/12 adversarial proposals rejected (100% intercepted), verifying the viability of cognitive agent-directed supervisory physical control in simulation.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on millisecond timescales via convex quadratic programming (QP), linear approximations degrade rapidly on highly non-linear chemical plants governed by exponential kinetics, such as Continuous Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, linearized controllers exhibit severe tracking error, control chattering, and valve saturation.

1.1 Related Work: Advanced & Learning-Based MPC

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018). Physics-informed neural network formulations (Raissi et al., 2019) have inspired domain-constrained residual observers that embed conservation laws into state estimation.

- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU and compute-shader architectures.

1.2 The Role of the Supervisory Layer: Lookahead vs. Reasoning

A critical insight in receding-horizon control is that numerical solvers optimize over a fixed finite horizon H_p (e.g., 4.0 s). When future disturbances are known in advance, mathematical solvers with preview can naturally pre-actuate within their horizon. However, in actual plant operations, forecasts arrive as human shift handover notes, laboratory feed analysis reports, or upstream DCS alarms. Classical MPC cannot parse these qualitative inputs.

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner interprets operator directives and contextual operational runbooks, translates them into quantitative cost manifolds, and verifies them through deterministic contract tests before any parameter update reaches physical actuators.

2. The Agentic MPC Architecture

2.1 The Two Halves: The Knower and The Doer

- **The Knower (Neural Reasoner):** Operates in *Intelligence Space*, translating operator directives into cost manifold parameters $\theta_{\mathcal{M}} = \{q_{\text{temp}}, q_{\text{barrier}}, u_{\text{target}}, \lambda\}$. Inference runs on dedicated cloud LPUs or server GPUs (e.g., Qwen-2.5-32B), with lightweight client-side WebGPU shaders (e.g., Qwen-2.5-0.5B via @mlc-ai/web-llm) available for offline edge operation.
- **The Doer (Executive Function Harness):** A deterministic verification kernel enforcing Ring 0 invariants (actuator saturation and slew rate limits), running automated contract unit tests, and operating preemptively below plant emergency shutdown thresholds.

2.2 The Four Structured Memory Subsystems

Memory Store	Storage Modality	Functional Role	CSTR Realization
The Playbook	Procedural (SKILL.md)	Verified operational runbooks with deterministic pre/post-conditions	cstr_runaway_mitigation.md verified via automated contract tests

Memory Store	Storage Modality	Functional Role	CSTR Realization
 The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0 \text{ K}$, nominal $T_{\text{ref}} = 350.0 \text{ K}$
 The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: V-101 → J-101 → CV-201 → M-101 → CH-3
 The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (MUTEX_PREC00L) preventing conflicting concurrent directives

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline (Jacobian QP)

The linear discrete-time optimal control problem is solved as a condensed Quadratic Program (QP) around the textbook steady state ($C_{A,\text{ref}} = 0.50 \text{ mol/L}$, $T_{\text{ref}} = 350.0 \text{ K}$, $T_{c,\text{base}} = 300.0 \text{ K}$):

$$\min_U \frac{1}{2} U^T H_{\text{qp}} U + g^T U \quad \text{subject to: } u_{\min} \leq u_k \leq u_{\max}, \quad |\Delta u_k| \leq \Delta u_{\max}$$

The continuous Jacobian matrix A has eigenvalues $[-0.0076, +0.0472] \text{ s}^{-1}$, exhibiting an open-loop thermal runaway pole ($\tau_{\text{growth}} \approx 21.2 \text{ s}$). State penalty is $Q = \text{diag}(10.0, 1.0)$, $R = 0.02$, input bounds $[280, 360] \text{ K}$, and slew limit 12 K/s .

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI evaluates $K = 1,024$ stochastic control rollouts sampled from $\mathcal{N}(0, \Sigma)$ with covariance $\Sigma = 6.0^2 I$ and temperature $\lambda = 10.0$. Discretized at rollout step $\Delta t_{\text{ctrl}} = 0.2 \text{ s}$:

$$x_{k+1} = f(x_k, v_k) + \Delta_{\text{PINN}}(x_k, v_k)$$

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (q_{\text{temp}} (T_k - T_{\text{ref}})^2 + q_{\text{barrier}} \max(0, T_k - T_{\text{barrier}})^2) + \frac{\lambda}{2} \sum_{k=0}^{H_p-1} \epsilon_k^T \Sigma^{-1} \epsilon_k$$

The optimal control update is computed via softmax importance weighting:

$$u_t^* = u_t + \sum_{m=1}^K w(U^{(m)}) \epsilon_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp\left(-\frac{1}{\lambda}(S(U^{(m)}) - \min_j S(U^{(j)}))\right)}{\sum_{k=1}^K \exp\left(-\frac{1}{\lambda}(S(U^{(k)}) - \min_j S(U^{(j)}))\right)}$$

3.3 Physics-Informed Residual Observer & UA Identification

An online observer continuously estimates the effective heat transfer coefficient $UA(t)$ at 10 Hz by minimizing the physics-informed thermal residual, identifying fouling degradation within 3.4 s of onset with <4.8% error:

$$\mathcal{L}_{\text{PINN}} = \left(\frac{dT}{dt} - \left[\frac{F}{V}(T_0 - T) + \frac{-\Delta H}{\rho C_p} k_0 e^{-E/RT} C_A - \frac{\widehat{UA}}{V \rho C_p} (T - T_c) \right] \right)^2$$

4. Benchmark Case Study: Exothermic CSTR

4.1 Governing Chemical Reactor Equations

$$\begin{aligned} \frac{dC_A}{dt} &= \frac{F}{V}(C_{A0} - C_A) - k_0 \exp\left(-\frac{E}{RT}\right) C_A \\ \frac{dT}{dt} &= \frac{F}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p} k_0 \exp\left(-\frac{E}{RT}\right) C_A - \frac{UA}{V \rho C_p} (T - T_c) \\ \frac{dT_c}{dt} &= \frac{1}{\tau_j}(u - T_c), \quad \text{with } |\dot{u}| \leq 12.0 \text{ K/s}, \quad u \in [280, 360] \text{ K} \end{aligned}$$

4.2 Benchmark Model Parameters & True Steady State

Parameter	Symbol	Nominal Value	Unit	Description
Reactor Volume	V	100.0	L	Vessel working volume
Volumetric Flow Rate	F	1.667	L/s	Feed throughput rate (100 L/min)
Feed Concentration	C_{A0}	1.0	mol/L	Inlet reactant concentration
Feed Temperature	T_0	350.0	K	Inlet feed stream temperature
Pre-exponential Factor	k_0	1.2×10^9	s^{-1}	Arrhenius frequency factor
Activation Energy Ratio	E/R	8,750.0	K	Arrhenius activation temperature
Heat of Reaction	$-\Delta H$	5.0×10^4	J/mol	Exothermic reaction enthalpy

Parameter	Symbol	Nominal Value	Unit	Description
Fluid Density	ρ	1,000.0	g/L	Reactor fluid density
Specific Heat Capacity	C_p	0.239	J/(g·K)	Reactor fluid heat capacity
Heat Transfer Area	UA	833.3	W/K	Nominal heat transfer coefficient
Jacket Thermal Lag	τ_j	2.0	s	Cooling jacket transport delay
Nominal Temperature	T_{ref}	350.0	K	Operating steady-state setpoint
Safe Operating Limit	T_{trip}	385.0	K	Plant emergency trip limit (SIS)
Preemptive Breaker	T_{breaker}	382.0	K	Software circuit breaker arming limit
Soft Barrier Threshold	T_{barrier}	380.0	K	Cost manifold barrier penalty start
Coolant Jacket Range	$T_{c,\text{min}}, T_{c,\text{max}}$	[280.0, 360.0]	K	Actuator saturation envelope
Jacket Slew Limit	$ \dot{T}_c _{\text{max}}$	12.0	K/s	15%/s of the 80 K span
Control Discretization	Δt_{ctrl}	0.20	s	Control step ($H_p = 20 \rightarrow 4.0$ s)
Plant Integration Step	Δt_{sim}	0.02	s	RK4 plant integration step (50 Hz)

4.3 Numerical Oracle Feasibility Benchmark

Dynamic trajectory optimization over the benchmark kinetic surge demonstrates the fundamental physical controllability limits:

Pre-Cooling Lead Time t_{lead}	Peak Reactor Temp T_{max} (K)	Emergency SIS Trip (≥ 385.0 K)	Operational Controllability Outcome
0.0 s (Reaction at Onset)	444.9 K	TRIP (Breached)	Unsurvivable runaway due to jacket lag ($\tau_j = 2$ s) & slew cap
1.0 s Lead Time	443.4 K	TRIP (Breached)	Insufficient thermal extraction ahead of exponential surge
2.0 s Lead Time	378.2 K	SAFE (Zero Trip)	Modulated dynamic pre-cooling contains surge below trip

Pre-Cooling Lead Time t_{lead}	Peak Reactor Temp T_{max} (K)	Emergency SIS Trip (≥ 385.0 K)	Operational Controllability Outcome
2.5 s Lead Time	367.8 K	SAFE (Zero Trip)	Stable containment with 17.2 K safety margin
3.0 s Lead Time	361.2 K	SAFE (Zero Trip)	Smooth containment
3.5 s Lead Time	356.5 K	SAFE (Zero Trip)	Robust containment (28.5 K margin to trip)
5.0 s Lead Time	350.0 K	SAFE (Zero Trip)	Fully absorbed thermal surge clamped at nominal setpoint

5. Verification & Safety Architecture

Agentic MPC enforces defense-in-depth: Ring 0 invariant checking ($q_T \geq 0.1$, $q_{\text{barrier}} \geq 1000$, $u \in [280, 360]$ K, $\text{slew} \leq 12$ K/s), automated contract unit tests on the active fouled plant model (<30 ms), a software preemptive breaker at 382 K, and an independent hardware SIS scram at 385 K (failsafe cooling valve full open, feed shut off).

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Benchmark Across 20 Randomized Seeds

Table 2 presents authentic simulation results across 20 randomized seeds computed directly from discrete QP, L-BFGS-B NMPC, and GPU MPPI path integral solvers:

Configuration	Controller Type	Surv RMSE (K)	All RMSE (K)	Peak T (K)	Slew (K/s)	SIS Trips	Settled
Baseline 1	Linear MPC (Jacobian QP, No Forecast)	2.56 ± 1.36	16.26 ± 5.52	389.3 ± 14.2	3.29 ± 0.95	12 / 20 (60.0%)	8 / 20 (9.8 s)
Baseline 2a	NMPC (L-BFGS-B, No Forecast)	2.48 ± 1.47	9.62 ± 5.37	372.9 ± 13.7	5.04 ± 1.29	6 / 20 (30.0%)	13 / 20 (6.9 s)
Ablation 1	MPPI (Static θ^* , No Forecast)	2.98 ± 1.75	12.14 ± 5.58	379.1 ± 14.6	2.34 ± 0.35	8 / 20 (40.0%)	11 / 20 (8.9 s)

Configuration	Controller Type	Surv RMSE (K)	All RMSE (K)	Peak T (K)	Slew (K/s)	SIS Trips	Settled
Ablation 2	MPPI + PINN (No Forecast)	2.60 ± 1.28	9.82 ± 5.40	373.5 ± 14.0	2.60 ± 0.36	6 / 20 (30.0%)	14 / 20 (12.1 s)
Baseline 2b	NMPC (With Forecast Preview)	1.47 ± 0.85	2.82 ± 2.93	356.7 ± 7.6	6.41 ± 1.03	1 / 20 (5.0%)	19 / 20 (2.6 s)
Ablation 3a	MPPI + PINN (With Forecast Preview)	1.81 ± 1.11	3.16 ± 3.00	357.3 ± 7.7	2.74 ± 0.23	1 / 20 (5.0%)	18 / 20 (5.8 s)
Ablation 3b	Rule Supervisor + MPPI	2.50 ± 1.66	8.03 ± 4.45	376.4 ± 15.5	3.80 ± 0.33	6 / 20 (30.0%)	12 / 20 (13.0 s)
Full System	Agentic MPC (Full Architecture)	0.95 ± 0.40	2.16 ± 2.56	355.4 ± 6.2	3.46 ± 0.26	1 / 20 (5.0%)	19 / 20 (2.8 s)

6.2 Key Scientific Insights

- 1. Preview is the Primary Physical Stabilizer:** Without preview, all controllers suffer trips in 30%–60% of seeds because jacket lag prevents reacting fast enough at onset. With 3.5 s preview, trip rates drop to 5% (1/20) across all nonlinear architectures.
- 2. Paired Difference & Quenching Avoidance:** Paired difference between MPPI+PINN Preview and Agentic Full yields mean difference $+0.999 \pm 0.764$ K ($t = 2.737, p = 0.0131$; Wilcoxon $W = 24.0, p = 0.0014$). Naive step switching (Rule Supervisor) caused 6 trips (30%) due to reaction quenching, whereas Agentic MPC's smooth pre-cooling ramp contained 19/20 seeds with 0.95 K non-trip RMSE.
- 3. The Supervisory Role of Agentic MPC:** The fundamental contribution is supervisory: translating qualitative shift notes and alarms into verified parameter manifolds, avoiding operator retuning, and providing formal contract gating.

6.3 Lead-Time Sensitivity Sweep

Lead Time	Trips	Mean Peak (K)	Max Peak (K)	Containment Status
0.0 s	9 / 10 (90%)	413.8 K	429.0 K	Severe thermal runaway
1.0 s	4 / 10 (40%)	379.9 K	418.4 K	Unstable transition

Lead Time	Trips	Mean Peak (K)	Max Peak (K)	Containment Status
2.0 s	3 / 10 (30%)	381.8 K	426.1 K	Marginally controllable
2.5 s	5 / 10 (50%)	388.9 K	429.2 K	Quenching sensitivity region
3.0 s	4 / 10 (40%)	382.2 K	424.3 K	Transition boundary
3.2 s	2 / 10 (20%)	372.0 K	425.3 K	Containment emerging
3.5 s	1 / 10 (10%)	358.2 K	418.4 K	Stable operational containment
4.0 s	1 / 10 (10%)	362.0 K	409.3 K	Robust containment
5.0 s	0 / 10 (0%)	350.8 K	362.1 K	Complete surge absorption

6.4 Evaluation of the Neural Reasoner & Confusion Matrix

Category	Description	Count	Accepted	Rejected	Outcome
Advisory Pre-cooling	Upstream surge alerts, fouling	12	12 (100%)	0 (0%)	Validated pre-cooling ramps
Routine Tracking	Setpoint, eco mode, damping	14	14 (100%)	0 (0%)	Standard MPC updates
Conservative Safety	Tightened envelope, high barrier	12	12 (100%)	0 (0%)	Elevated safety parameters
Adversarial Proposals	Disable barrier, out-of-bounds u	12	0 (0%)	12 (100%)	Intercepted deterministically
Total Test Suite	Comprehensive 50-Directive Suite	50	38 (76%)	12 (24%)	100% Gating Accuracy (Zero Leaks)

7. Industrial Deployment Limits & Conclusion

While Agentic MPC demonstrates robust supervisory capabilities in simulation, certified industrial deployment requires strict separation of concerns. Under IEC 61511 / ISA-84, safety instrumented functions must remain physically independent, deterministic, and SIL-rated. The agentic layer operates solely in the basic process control system (BPCS) supervisory space and must never share hardware, sensors, or actuators with the emergency shutdown interlock.

Systematic ablation across eight control configurations demonstrates that while numerical preview lookahead provides the essential physical mechanism of thermal stabilization, the supervisory agentic layer provides critical directive translation, formal contract gating, and smooth pre-cooling modulation that eliminates the hazardous quenching trips of naive rule-based switching. Automated contract testing and Ring 0 invariant verification provide the necessary deterministic gating to transition agentic control systems toward industrial deployment.

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